FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

FIITJEE **JEE (Main), 2016**

ALL INDIA INTEGRATED TEST SERIES

PART TEST – IV

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

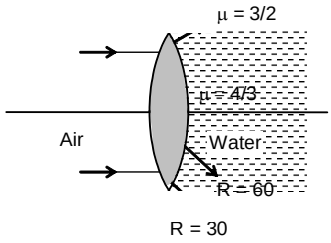
PART – I

SECTION – A

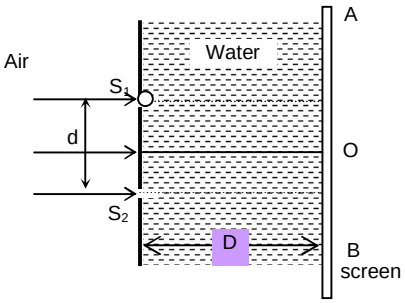
(Single Correct Choice Type)

This section contains ***30 multiple choice questions***. Each question has four choice (A), (B), (C) and (D) out of which ***ONLY ONE*** is correct.

1. A thin lens with radius of curvature 30 cm and 60 cm is placed with water on left side and air on right side, then the focal length of the lens for the parallel incident rays shown in the figure, is
(A) 72 cm (B) 144 cm
(C) 36 cm (D) none of these



2. Young's double slit experiment is submerged in water as shown in the figure. If an air bubble is formed on slit S_1, then the fringe pattern
(A) will shift towards A.
(B) will shift towards B.
(C) will not shift.
(D) can't say



3. A screw gauge used to determine the diameter of a wire has accuracy of the order of 0.001 cm. The same screw gauge is used to determine the radius of a wire as well as its length. If the length of the wire is found to be 2 cm and diameter is determined to be 0.5mm, the maximum percentage error accumulated in the resistance of the wire is
(A) 3.05 % (B) 3.1 %
(C) 4.05 % (D) 2.05 %

Space for rough work

4. A light of wavelength λ_1 is incident on a photoelectric plate. It is found that for certain potential difference the no. of photoelectrons reaching the collector plate is zero. When light of wavelength λ_2 is used, the no. of photoelectrons reaching the collector plate is zero at twice the above value of potential difference. The threshold wavelength for the photoelectric surface is

(A) $\frac{\lambda_1\lambda_2}{\lambda_2 - 2\lambda_1}$ (B) $\frac{2\lambda_1\lambda_2}{\lambda_2 - \lambda_1}$

(C) $\frac{\lambda_1\lambda_2}{\lambda_2 - \lambda_1}$ (D) $\frac{\lambda_1\lambda_2}{2\lambda_2 - \lambda_1}$

5. Consider two resistances R_1 $(5 \pm 0.2)\ \Omega$ and $R_2 = (10 \pm 0.1)\ \Omega$. Find the ratio of their equivalent resistances when they are connected in series and in parallel, with proper error limits.

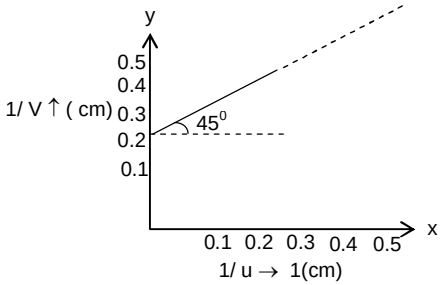
(A) (15 ± 0.5) (B) (4.5 ± 0.4)

(C) (15 ± 0.3) (D) (4.5 ± 0.5)

6. An experiment was conducted in the laboratory using a thin converging lens. The reciprocal of object distance (u) vs image distance (v) was plotted as shown. For the graph, the focal length of the lens used for the experiment, is

(A) 10 cm (B) 7.5 cm

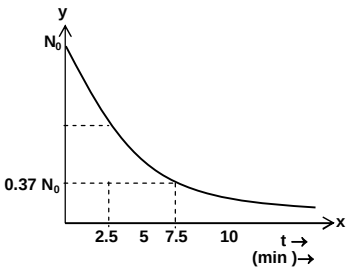
(C) 5 cm (D) 0.5 cm



7. A radioactive sample undergoes decay as shown in the graph. Calculate the amount of sample left after 30 min. if initial amount of nuclei present is N_0, is

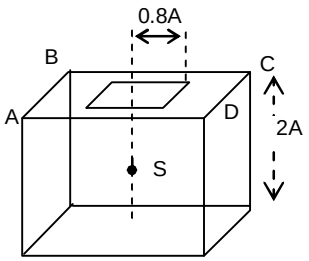
(A) N_0 / e^6 (B) N_0 / e^4

(C) N_0 / e^3 (D) N_0 / e^2



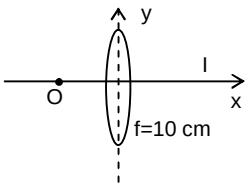
Space for rough work

8. A cube of side 2A has a point source S at its centre. All the faces of the cube are blackened except the top surface which has got a square sheet of black paper of side 1.6 A. If the source starts oscillating inside a negligibly thin tunnel perpendicular to the direction of face ABCD, and having the mean position of oscillation at the centre of the ABC, then the maximum amplitude of oscillation so that the source may not be seen from the top surface is (Refractive index of material of cube $\mu = \sqrt{3.25}$. Assume black surface absorbs all light incident on it.)



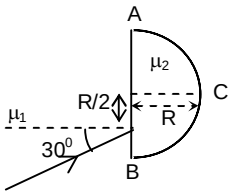
(A) 0.1 A (B) 0.2 A
(C) 0.3 A (D) 0.4 A

9. An object is placed at a distance of 15 cm from converging lens of focal length 10 cm. Now if lens is displaced laterally (along y – axis) by 1 cm, the lateral shift of image will be



(A) 1 cm (B) 2 cm
(C) 3 cm (D) 4 cm.

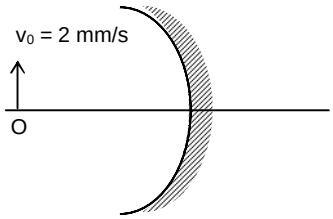
10. A ray of light travelling in air (μ_1 = 1) strikes the surface AB of semicylindrical surface at an angle 30^0 from normal to the surface. If ray come out of slab parallel to itself from C, the refractive index (μ_2) of slab will be



(A) $\mu_2 = \frac{\sqrt{5}}{2}$ (B) $\mu_2 = \frac{\sqrt{5}}{\sqrt{2}}$

(C) $\frac{2\sqrt{5}}{3}$ (D) $\frac{4}{3}$

11. A point object placed at a distance of 15 cm in front of concave mirror of radius of curvature 20 cm. If object starts moving laterally at a speed of v_0 = 2 mm/s. as shown in the figure the lateral speed of image will be



(A) v = 1 mm/s (B) v = 2 mm/s
(C) v = 4 mm/s (D) none of these

Space for rough work

12. In Young's double slit experiment the maximum intensity obtained is I_0. The intensity at a point on screen where the phase difference between two interfering waves is $\pi/3$ is
(A) $I_0/4$ (B) $4I_0$
(C) I_0 (D) $\frac{3I_0}{4}$

13. An electron jumps from 7th orbit to 2nd orbit in an excited hydrogen like atom, the number of lines observed in the resulting spectrum is
(A) 10 (B) 15
(C) 12 (D) 8

14. A radionuclide Z is produced at constant rate k and decays with decay constant 'λ'. The maximum number of nuclei of radionuclide Z, (N_{max}) if number of nuclei of z are zero at t = 0,
(A) $N_{max} = \frac{\lambda}{k}$ (B) $N_{max} = \frac{k}{2\lambda}$
(C) $N_{max} = \frac{2k}{\lambda}$ (D) $N_{max} = \frac{k}{\lambda}$

15. A light of wavelength 280 nm is used in an experiment on photoelectric surface of lithium (ϕ = 2.5 eV) as cathode. The kinetic energy of electron coming out of cathodes cannot be equal to
(A) 1 eV (B) zero
(C) 2.4 eV (D) 1.9 eV

16. An element ${}_ZX^A$ undergoes n α - emissions followed by m β - emissions. If element formed is ${}_{z-2}X^{A-8}$, then
(A) n = 2, m = 4 (B) n = 4, m = 4
(C) n = 2, m = 2 (D) n = 4, m = 2

17. In Young's double slit experiment the angular fringe width for given light is $\Delta\theta$. Now if wavelength of light is doubled while distance between slit is halved, the angular fringe width will become
(A) $\frac{\Delta\theta}{2}$ (B) $\Delta\theta$
(C) $4\Delta\theta$ (D) $\frac{\Delta\theta}{4}$

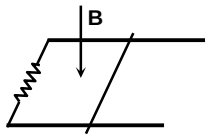
Space for rough work

18. The relation between the wavelength of radiation (λ) and the principal quantum no. n, when an electron jumps from $(n-1)^{th}$ orbit to n^{th} orbit [Assume n is large]

(A) $\lambda \propto n$ (B) $\lambda \propto n^2$

(C) $\lambda \propto n^3$ (D) $\lambda \propto n^4$

19. Two parallel rails of negligible resistances are fixed on a horizontal surface. One end of the rails are connected by a resistance R as shown in the figure. There is a uniform magnetic field B perpendicular to the plane of the rails. A sliding wire of negligible resistance, length ℓ and mass m can slide on the rails. The coefficient of friction between the rails and wire is μ. The minimum force required to move the wire with uniform velocity v is



(A) $\mu mg + \frac{B^2\ell^2}{R}v$ (B) $\frac{B^2\ell^2}{R}v$

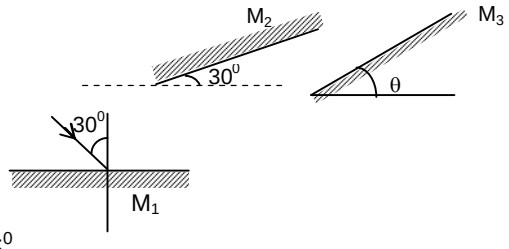
(C) $\frac{\mu mg}{\sqrt{\mu^2+1}} + \frac{B^2\ell^2}{R}\frac{v}{\sqrt{\mu^2+1}}$ (D) None of these

20. The length of a strip measured with metre rod is 10.0 cm. Its width measured with verniers callipers is 1.00 cm. The least count of metre rod is 0.1 cm and that of vernier callipers is 0.01 cm. the error in the measurement of its area is

(A) $\pm$ 0.01 cm^2 (B) $\pm$ 0.1 cm^2

(C) $\pm$ 0.11 cm^2 (D) $\pm$ 0.2 cm^2

21. M_1, M_2 and M_3 are plane mirror arranged as shown in the figure. A ray of light is incident on the mirror M_1 at an angle of 30^0 with the normal. It is observed that the ray of light after suffering reflection with M_1 followed by M_2 and M_3 emerges vertically upwards. The angle of inclination of mirror M_3 if M_2 is kept inclined at 30^0 to the horizontal is

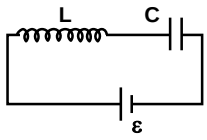


(A) 25^0 (B) 35^0

(C) 45^0 (D) 30^0

Space for rough work

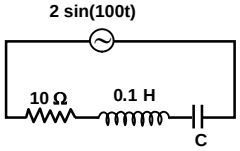
22. An inductor L and uncharged capacitor C is connected to a cell of emf ε as shown in the figure. At t = 0, the switch is closed. Find charge on the capacitor q as a function of time t.



(A) $q = C\varepsilon \sin\left(\sqrt{\frac{1}{LC}}t\right)$ (B) $q = C\varepsilon \cos\left(\sqrt{\frac{1}{LC}}t\right)$

(C) $q = C\varepsilon\left[1-\cos\sqrt{\frac{1}{LC}}t\right]$ (D) $q = C\varepsilon\left[1-\sin\sqrt{\frac{1}{LC}}t\right]$

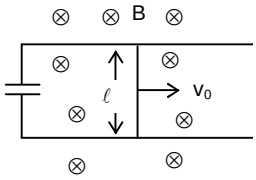
23. The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to

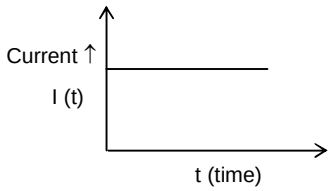


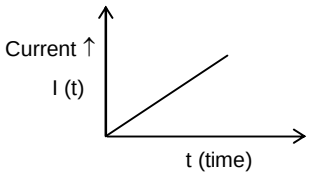
(A) 400 μF (B) 300 μF

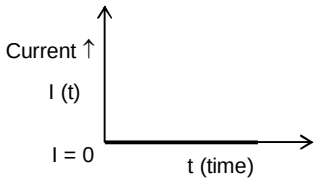
(C) 500 μF (D) 200 μF

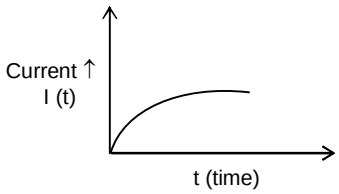
24. Two infinitely long conducting parallel rails are connected through a capacitor of capacitance C as shown in the figure. A perfect conductor of length ℓ is moved with constant speed v_0. Which of the following graph truly depicts the variation of current *through* the conductor with time ?



(A)


(B)


(C)


(D)


Space for rough work

25. Light from a monochromatic source is incident normally on a small photo sensitive surface having surface area S and work function ϕ. If power of source is k and a is the distance between source and the surface then

(i) number of photons striking the surface per unit time will be $\frac{k\lambda S}{4\pi hca^2}$

(ii) maximum energy of the emitted electrons will be $\frac{1}{\lambda}(hc - \lambda\phi)$

(iii) stopping potential needed to stop the most energetic photoelectrons will be $\frac{e}{\lambda}(hc - \lambda\phi)$

(iv) photo emission occurs only if $0 < \lambda \leq \frac{hc}{\phi}$

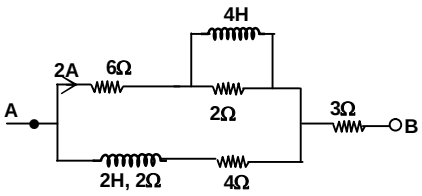
(A) (i) and (ii) are true.
(B) (ii) and (iii) are true.
(C) (i), (ii) and (iv) are true.
(D) (i), (ii) and (iii) are true.

26. If the potential energy of proton & electron of a hypothetical hydrogen like atom is given by $U = -\frac{he^2}{2r}$ where e = charge of electron and r is the radius of atom, then the radius of Bohr's first orbit is given by (Mass of electron is m_e)

(A) $\frac{h^2}{2\pi^2 me^2}$
(B) $\frac{h}{2\pi^2 me^2}$
(C) $\frac{e^2h^2}{2\pi^2 m}$
(D) $\frac{h}{4\pi^2 me^2}$

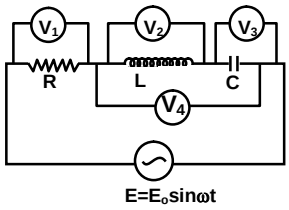
27. Find $V_A - V_B$ in steady state in figure shown below

(A) 8 V
(B) 24 V
(C) 16 V
(D) none



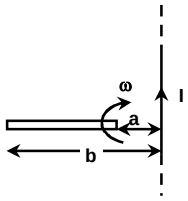
Space for rough work

28. In the given figure, which voltmeter will read zero voltage at resonant frequency ω rad/sec?



(A) V_1 (B) V_2
(C) V_3 (D) V_4

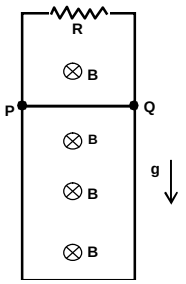
29. A copper rod moves with a constant angular velocity ω, about a long straight wire carrying a current I. If the ends of the rod from the wire are at distances a and b, then the e.m.f. induced in the rod is



(A) $\frac{\mu_0 i(\omega a)}{2\pi}\ln\left(\frac{b}{a}\right)$ (B) $\frac{\mu_0 i(\omega b)}{2\pi}\ln\left(\frac{b}{a}\right)$

(C) zero (D) $\frac{\mu_0 i\omega(a+b)}{4\pi}\ln\left(\frac{b}{a}\right)$

30. A rod PQ mass 'm' and length ℓ can slide without friction on two vertical conducting semi infinite rails. It is given a velocity V_0 downwards, so that it continues to move downward with the same speed V_0 on its own at any later instant of time. Assuming g to be constant every where, the value of $\frac{V_0B^2L^2}{mgR}$ is



(A) 1 (B) 2
(C) 3 (D) 4

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only one is correct.

1. A mixture contains 1 mole of volatile liquid A (P_A° = 100 mm Hg) and 3 moles of volatile liquid B(P_B° = 80 mm Hg). If solution behaves ideally, the total vapour pressure of the distillate is
   (A) 85 mm Hg (B) 85.88 mm Hg
   (C) 90 mm Hg (D) 92 mm Hg

2. 0.04 M Na_2SO_4 solution is isotonic with 0.1 M glucose at the same temperature. What is the apparent degree of dissociation of Na_2SO_4?
   (A) 0.25 (B) 0.50
   (C) 0.75 (D) 0.85

3. When mercuric iodide is added to the aqueous solution of potassium iodide, the
   (A) freezing point is raised. (B) freezing point does not change.
   (C) freezing point is lowered. (D) boiling point does not change.

4. A 0.001 molal solution of $Pt(NH_3)_4Cl_4$ in water had a freezing point depression of 0.0054°C. If K_f for water is 1.80, the correct formula for the above compound assuming its complete dissociation is
   (A) $[Pt(NH_3)_4Cl_3]Cl$ (B) $[Pt(NH_3)_4Cl_2]Cl_2$
   (C) $[Pt(NH_3)_4Cl]Cl_3$ (D) $[Pt(NH_3)_4Cl_4]$

5. Yellow ammonium sulphide is a suitable reagent for the separation of
   (A) Bi_2S_3 and CuS (B) PbS and Bi_2S_3
   (C) HgS and PbS (D) CdS and As_2S_3

Space for rough work

6. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolved on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is
(A) Hg_2^{2+} salt (B) Cu^{2+} salt
(C) Ag^{+} salt (D) Pb^{2+} salt

7. Ammonium compound on heating produced a gas which gives white fumes when a rod dipped in HCl is brought in contact with the gas. Which one of the following will not produce the gas on heating?
(A) $(NH_4)_2SO_4$ (B) $(NH_4)_2SO_3$
(C) NH_4NO_2 (D) NH_4Cl

8. $H_2C_2O_4 \xrightarrow{H_2SO_4} A(gas) + B(gas) + C(liquid)$ oxalic acid
Gas (A) burns with blue flame and is oxidized to gas (B). Gas (B) turns lime water milky
(A), (B) & (C) are respectively
(A) CO_2, CO, H_2O (B) CO, CO_2, H_2O
(C) CO, SO_2, H_2O (D) SO_2, CO, H_2O

9. A solid compound AB has NaCl structure. If the radius of the cation is 100 pm, the radius of anion 'B' is
(A) 241.5 pm (B) 141.5 pm
(C) 41.5 pm (D) 21.5 pm

10. 0.1 M $Cr_2O_7^{2-}$ will be a better oxidizing agent in which of the following solution?
(A) 10 ml of 1 M CH_3COOH + 10 ml of 1 M NH_4OH
(B) 100 ml of 0.1 M NaOH + 100 ml of 0.1 M HCl
(C) 50 ml of 0.1 M HCl
(D) 100 ml of 10^{-7} M HCl

11. A cell made up of two hydrogen electrodes. The positive electrode is in contact with a solution of 10^{-4} M hydrogen ions. The emf of the cell is 0.118 V at 25°C. The concentration of hydrogen ions at negative electrode is
(A) 10^{-6} M (B) 10^{-4} M
(C) 10^{-2} M (D) None of the above

Space for rough work

12. One litre of 1 M $CuSO_4$ solution is electrolysed using copper and inert electrodes respectively. After passing 2F of electricity, molarity of $CuSO_4$ respectively, will be

(A) $\frac{M}{2}, M$ (B) $\frac{M}{4}, 0$

(C) $M, \frac{M}{2}$ (D) M, 0

13. Equilibrium constant for the dissociation of $Ag(NH_3)_2^+$ into Ag^+ and NH_3 is 6×10^{-8}. Calculate E^o for the following half reaction, $Ag(NH_3)_2^+ + e^- \longrightarrow Ag(s) + 2NH_3$

($E^o_{Ag^+/Ag} = 0.799$ V, log 6 = 0.7781).

(A) 0.372 (B) 0.799

(C) 0.7781 (D) 0.6

14. The composition of a sample of Wustite is $Fe_{0.93}O$. What is the percentage of iron present as Fe^{3+}?

(A) 15% (B) 25%

(C) 35% (D) 45%

15. In a compound AB the ionic radii of A^+ and B^- are 88 pm and 200 pm respectively the coordination number of A^+ is

(A) 6 (B) 8

(C) 4 (D) 12

16. Radius ratio $\frac{r^+}{r^-} = \frac{88}{200} = 0.44$. This corresponds to co-ordiation number 6. 20. Bridging bonds in diborane are

(A) 3 centre – 4 electrons (B) 2 centre – 3 electrons

(C) 3 centre – 2 electrons (D) 3 centre – 3 electrons

17. The incorrect statement regarding ClO_2 molecule is that

(A) it contains odd number of electrons and hence paramagnetic.

(B) ClO_2 dimerises.

(C) the Cl – O bond length is shorter than the single Cl – O bond length.

(D) it is a highly reactive species.

Space for rough work

18. Aqueous sodium hydroxide reacts with white phosphorus to form phosphine and
(A) NaH_2PO_3 (B) P_2O_5
(C) Na_3PO_3 (D) P_2O_3

19. Pick out the incorrect statement with respect to SO_3
(A) In solid state SO_3 exists as cyclic trimer in which four oxygen atoms are arranged approximately tetrahedrally around each S atom
(B) In presence of moisture cyclic trimer changes into linear polymerized structure
(C) It reacts explosively with H_2O to form H_2SO_4
(D) It does not react with HBr

20. Van Arkel method of purification of metals involves converting the metal to a:
(A) volatile stable compound. (B) non-volatile stable compound.
(C) volatile unstable compound. (D) none of these.

21. The ore carnallite is represented by structure
(A) Na_3AlF_6 (B) $Na_2Al_2O_3$
(C) Fe_3O_4 (D) $KCl.MgCl_2.6H_2O$

22. Auto reduction process is used for the extraction of
(A) Hg (B) Pb
(C) Cu (D) All of these

23. Pb and Sn are extracted from their chief ore respectively by
(A) Carbon reduction and self reduction (B) Self reduction and carbon reduction
(C) Electrolysis and self reduction (D) Self reduction and electrolysis

24. Pink colour of $KMnO_4$ is due to
(A) d – d transition (B) metal to ligand transition
(C) ligand to metal transition (D) All the above

Space for rough work

25. In which compound does vanadium have on oxidation number of +4?
(A) NH_4VO_2 (B) $K_4[V(CN)_6]$
(C) VSO_4 (D) $VOSO_4$

26. The yellow colour solution of Na_2CrO_4 changes to orange red on passing CO_2 gas due to the formation of
(A) CrO_5 (B) CrO_3
(C) $Na_2Cr_2O_7$ (D) Cr_2O_3

27. Considering H_2O as a weak field ligand, the number of unpaired electrons in $[Mn(H_2O)_6]^{2+}$ will be
(A) Two (B) Four
(C) Three (D) Five

28. The smog is essentially caused by presence of
(A) O_2 and O_3 (B) O_3 and N_2
(C) oxides of sulphur and nitrogen (D) O_2 and N_2

29. Which molecule has tetrahedral geometry?
(A) $[Co(NH_3)_6]^{3+}$ (B) $[Ni(CN)_4]^{2+}$
(C) $Fe(CO)_5$ (D) $[NiCl_4]^{2-}$

30. Which of the following is wrong in the given statements?
(A) Coordination number of Cl^- in CsCl structure is 8.
(B) Ca^{+2} is present in tetrahedral voids of the cubic closed packed structure of F^-
(C) O^{-2} is present in the body centre of cube formed by Na^+ ions
(D) In NaCl schottky defect are observed.

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The vectors $\vec{a},\vec{b}$ and $\vec{c}$ are of the same length and taken pairwise, they form equal angles. If $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=\hat{j}+\hat{k}$, then the co-ordinates of $\vec{c}$ are

(A) (1, 0, 1) (B) (1, 2, 3)
(C) (–1, 1, 2) (D) None of these

2. The value of c so that for all real x, the vectors $cx\hat{i}-6\hat{j}+3\hat{k}$, $x\hat{i}+2\hat{j}+2cx\hat{k}$ make an obtuse angle are

(A) $c<0$ (B) $0<c<\frac{4}{3}$
(C) $-\frac{4}{3}<c<0$ (D) $c>0$

3. $\vec{a},\vec{b},\vec{c}$ are three non-zero, non-coplanar vectors and $\vec{p},\vec{q},\vec{r}$ are three other vectors such that $\vec{p}=\frac{\vec{b}\times\vec{c}}{\vec{a}.\vec{b}\times\vec{c}}$, $\vec{q}=\frac{\vec{c}\times\vec{a}}{\vec{a}.\vec{b}\times\vec{c}}$, $\vec{r}=\frac{\vec{a}\times\vec{b}}{\vec{a}.\vec{b}\times\vec{c}}$. Then $[\vec{p}\,\vec{q}\,\vec{r}]$ equals

(A) $\vec{a}\,.\,\vec{b}\times\vec{c}$ (B) $\frac{1}{\vec{a}.\vec{b}\times\vec{c}}$
(C) 0 (D) None of these

Space for rough work

4. If S is the circumcentre, G the centroid, O the orthocentre of a triangle ABC, then $\overrightarrow{SA}+\overrightarrow{SB}+\overrightarrow{SC}=$

(A) $\overrightarrow{SG}$ (B) $\overrightarrow{OS}$

(C) $\overrightarrow{SO}$ (D) $\overrightarrow{OG}$

5. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors such that volume of parallelopiped formed with $\vec{a},\vec{b},\vec{c}$ as coterminous edges is equal to volume of parallelepiped formed with $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ as coterminous edges then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ may be

(A) $[\vec{a}\vec{b}\vec{c}]=0$ (B) $[\vec{a}\vec{b}\vec{c}]=1$

(C) $[\vec{a}\vec{b}\vec{c}]=-2$ (D) $[\vec{a}\vec{b}\vec{c}]=2$

6. If $\vec{a}$ and $\vec{b}$ are two unit vectors, then the vector $\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)$ is parallel to the vector

(A) $\vec{a}-\vec{b}$ (B) $\vec{a}+\vec{b}$

(C) $2\vec{a}-\vec{b}$ (D) $2\vec{a}+\vec{b}$

7. The point of intersection of the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ is

(A) $\left(\frac{1}{2},\frac{1}{2},-\frac{3}{2}\right)$ (B) $\left(-\frac{1}{2},-\frac{1}{2},\frac{3}{2}\right)$

(C) $\left(\frac{1}{2},-\frac{1}{2},-\frac{3}{2}\right)$ (D) $\left(-\frac{1}{2},\frac{1}{2},\frac{3}{2}\right)$

8. The angle between the line $\frac{x+1}{3}=\frac{y-1}{2}=\frac{z-2}{4}$ and the plane $2x+y-3z+4=0$, is

(A) $\sin^{-1}\left(\frac{4}{\sqrt{406}}\right)$ (B) $\sin^{-1}\left(\frac{-4}{\sqrt{406}}\right)$

(C) $\sin^{-1}\left(\frac{4}{14\sqrt{229}}\right)$ (D) None of these

Space for rough work

9. The equation of the plane in which the lines $\frac{x-5}{4}=\frac{y-7}{4}=\frac{z+3}{-5}$ and $\frac{x-8}{7}=\frac{y-4}{1}=\frac{z-5}{3}$ lie, is

(A) $17x-47y-24z+172=0$ (B) $17x+47y-24z+172=0$

(C) $17x+47y+24z+172=0$ (D) $17x-47y+24z+172=0$

10. The direction ratios of the lines OA and OB are 1, –2, –1 and 3, –2, 3. Then the direction cosines of the normal of plane AOB where O is the origin, are

(A) $\left(\frac{4}{\sqrt{29}},\frac{3}{\sqrt{29}},\frac{-2}{\sqrt{29}}\right)$ (B) $\left(\frac{2}{\sqrt{29}},-\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}\right)$

(C) $\left(-\frac{4}{\sqrt{29}},\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}}\right)$ (D) $\left(\frac{4}{\sqrt{29}},\frac{2}{\sqrt{29}},\frac{-3}{\sqrt{29}}\right)$

11. The length of projection, of the line segment joining the points (1, –1, 0) and (–1, 0, 1), to the plane 2x + y + 6z = 1, is equal to

(A) $\sqrt{\frac{255}{61}}$ (B) $\sqrt{\frac{237}{61}}$

(C) $\sqrt{\frac{137}{61}}$ (D) $\sqrt{\frac{155}{61}}$

12. Reflection of the line $\frac{x-1}{-1}=\frac{y-2}{3}=\frac{z-4}{1}$ in the plane x + y + z = 7 is

(A) $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-4}{1}$ (B) $\frac{x-1}{-3}=\frac{y-2}{-1}=\frac{z-4}{1}$

(C) $\frac{x-1}{-3}=\frac{y-2}{1}=\frac{z-4}{-1}$ (D) $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z+4}{1}$

13. A man alternately tosses a coin and throws a dice beginning with the coin. The probability that he gets a head in the coin before he gets a 5 or 6 in the dice is

(A) 3/4 (B) 1/2

(C) 1/3 (D) None of these

Space for rough work

14. Let p denotes the probability that a man aged x years will die in a year. The probability that out of n men $A_1, A_2, A_3 \ldots\ldots A_n$ each aged x, A_1 will die in a year and will be the first to die,

(A) $\frac{1}{n}[1-(1-p)^n]$ (B) $[1-(1-p)^n]$

(C) $\frac{1}{n-1}[1-(1-p)^n]$ (D) None of these

15. One hundred identical coins each with probability p of showing up heads are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, then the value of p is

(A) $\frac{1}{2}$ (B) $\frac{49}{101}$

(C) $\frac{50}{101}$ (D) $\frac{51}{101}$

16. A man takes a step forward with probability 0.4 and backward with probability 0.6. The probability that at the end of eleven steps he is one step away from the starting point is

(A) ${}^{11}C_6 (0.24)^5$ (B) ${}^{11}C_6 (0.4)^6 (0.6)^5$

(C) ${}^{11}C_6 (0.6)^6 (0.4)^5$ (D) None of these

17. A random variable X has Poisson's distribution with mean 2. Then $P(X > 1.5)$ equals

(A) $1-\frac{3}{e^2}$ (B) $\frac{3}{e^2}$

(C) $\frac{2}{e^2}$ (D) 0

Space for rough work

18. If $1, \omega, \omega^2$ are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to

(A) 0 (B) 1

(C) ω (D) ω^2

19. Matrix A is such that $A^2 = 2A - I$, where I is the identity matrix. Then for $n \geq 2, A^n =$

(A) $nA - (n-1)I$ (B) $nA - I$

(C) $2^{n-1}A - (n-1)I$ (D) $2^{n-1}A - I$

20. If $a_1, a_2, a_3 ..... a_n ....$ are in G.P. then the value of the determinant $\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$ is

(A) – 2 (B) 1

(C) 2 (D) 0

21. If value of a third order determinant is 11, then the value of the square of the determinant formed by the cofactors will be

(A) 11 (B) 121

(C) 1331 (D) 14641

22. If $F(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $G(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$, then $[F(\alpha)G(\beta)]^{-1} =$

(A) $F(\alpha) - G(\beta)$ (B) $-F(\alpha) - G(\beta)$

(C) $[F(\alpha)]^{-1}[G(\beta)]^{-1}$ (D) $[G(\beta)]^{-1}[F(\alpha)]^{-1}$

Space for rough work

23. For how many value (s) of x in the closed interval $[-4,-1]$ is the matrix $\begin{bmatrix} 3 & -1+x & 2 \\ 3 & -1 & x+2 \\ x+3 & -1 & 2 \end{bmatrix}$ singular

(A) 2 (B) 0

(C) 3 (D) 1

24. The area bounded by $x = x_1$, $y = y_1$ and $y = -(x+1)^2$ where x_1, y_1 are the values of x, y satisfying the equation $\sin^{-1}x + \sin^{-1}y = -\pi$ will be, (nearer to origin).

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 1 (D) $\frac{3}{2}$

25. Let $f(x) = x + \sin x$. The area bounded by $y = f^{-1}(x), y = x, x \in [0,\pi]$ is

(A) 1 (B) 2

(C) 3 (D) canot be found because $f^{-1}(x)$ cannot be determined

26. The area of the smaller region in which the curve $y = \left[\frac{x^3}{100} + \frac{x}{50}\right]$ where [.] denotes the greatest integer function, divides the circle $(x-2)^2 + (y+1)^2 = 4$, is equal to

(A) $\frac{2\pi - 3\sqrt{3}}{3}$ sq. units (B) $\frac{3\sqrt{3} - \pi}{3}$ sq. units

(C) $\frac{4\pi - 3\sqrt{3}}{3}$ sq. units (D) $\frac{5\pi - 3\sqrt{3}}{3}$ sq. units

Space for rough work

27. Let f(x) be a continuous function such that the area bounded by the curve y = f(x), the x-axis and the two ordinates x = 0 and x = a is $\frac{a^2}{2}+\frac{a}{2}\sin a+\frac{\pi}{2}\cos a$ then $f\left(\frac{\pi}{2}\right)$ is

(A) $\frac{1}{2}$ (B) $\frac{\pi^2}{8}+\frac{\pi}{4}$

(C) $\frac{\pi+1}{2}$ (D) $\frac{\pi^2}{8}-\frac{\pi}{4}$

28. The area bounded by the curve $y=\sqrt{\sin[x]+[\sin x]}$, lines $x=1, x=\frac{4}{3}$ and x-axis is

(A) $\sqrt{\sin 1}$ Sq.units (B) $\frac{1}{3}\sqrt{\sin 1}$ Sq.units

(C) $\frac{4}{3}\sqrt{\sin 1}$ Sq.units (D) None of these

29. Let $f:[0,\infty)\to R$ be a continuous and strictly increasing function such that $f^3(x)=\int_0^x t.f^2(t)\,dt, \forall x>0$. The area enclosed by $y=f(x)$, the x-axis and the ordinate at x=3, is

(A) 1 (B) $\frac{3}{2}$

(C) 2 (D) 3

30. The order of the differential equation whose general solution is given by $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}$, where c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is

(A) 5 (B) 4

(C) 3 (D) 2

Space for rough work